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GMBA 5885 W01
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Game Theory and Strategic Thinking

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Opening

- What is the difference between the following two decisions?

  1) *I am looking for the best cinema in Wenzhou for tonight's movie.*

  2) *I am looking for the best cinema in Wenzhou,* ***yet avoiding my ex-girlfriend***.

- In a game situation, The final outcome is often influenced not only by your decisions but also by others', yet the latter is not under your direct control.

  - **Interdependence** is the key aspect of game situation and **strategic thinking** is the focus of game theory.

Strategic reasoning is everything!!

Opening – Is This A Game?

Situation	Game?	Why
Price competition between Coca-Cola and Pepsi	?	Each firm's pricing decision affects the other's profit
Two students deciding whether to study or free-ride in a group project	?	Each student's payoff depends on the other's effort
Negotiating salary with a job candidate	?	Each side responds strategically to the other's demands
Choosing whether to carry an umbrella based only on weather forecast	?	Outcome depends on nature, not another strategic player
Rolling a dice in a casino	?	Chance matters more than strategic interaction

Opening – cont.

- **Game theory** was developed in the early 1940's (in connection with war strategies during the cold war era) and is concerned with the formal study of rational decisions in 'strategic' situations where conflict interfaces with mutual dependence. Such situations are common both within firms and in market actions among firms as well as in everyday life.

  – price wars (e.g. collusion vs competition?) or trade wars (e.g. managed trade vs free trade)

- In these situations, players have their own objectives – possibly in conflict - but also share a mutual dependence.

Motivating Stories

Example: Location Game – Picking Locations on a Beach Road

- Two street vendors, say **A** and **B**, on a linear beach road, trying to find proper locations for their shops.
- Their customers are uniformly spread out all through the beach and are in general averse to walking.

- ❑ Horizontal: price
- ❑ Vertical: quality

Challenge: Where should the vendors locate their shops? What is the logic?

Fundamentals of Games

A Game is composed of the following elements:

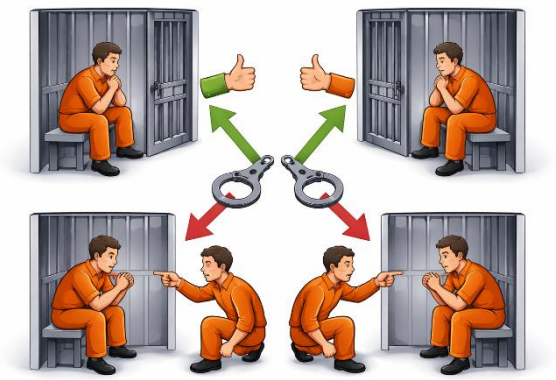
- **Players**: A set of rational, payoff-maximizing competitors
  - firms, governments or any other decision groups
- **Strategies**: Set of actions / choices players can select as a game plan
  - Prices/outputs/investments, tariff rates, threats/promises etc.
- **Payoffs/preferences**: Value associated with each possible outcome of the game
  - (expected) economic profits, utility or national welfare etc.

Regarding the payoffs, it is not the absolute size that matters. Ordering matters. Payoffs are ordinal not cardinal.

Example: Prisoner's Dilemma

Prisoner B / Prisoner A	Prisoner B stays silent (*cooperates*)	Prisoner B testifies (*defects*)
Prisoner A stays silent (*cooperates*)	Each serves 1 year	Prisoner A: 3 years Prisoner B: goes free
Prisoner A testifies (*defects*)	Prisoner A: goes free Prisoner B: 3 years	Each serves 2 years

		Player B	
		Deny	Confess
Player A	Deny	(2, 2)	(0, 3)
	Confess	(3, 0)	(1, 1)



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- Two players
- Strategies or actions
- Preference

Cooperative vs. Non-cooperative Games

- In a non-cooperative game, each player is to make her own decision independently. Thus "explicit" collusion is not allowed.
- In a cooperative game, all or groups of players ("coalitions") are allowed to enforce cooperative and collusive behavior, hence the game is often a competition among coalitions of players, rather than among individual players. (Example: trade agreement negotiations)

- In business, collusion is attractive but illegal.

  In a corporate world, only the games of non-cooperative settings matter hence we will deal with non-cooperative games only.
- Therefore, we only study non-cooperative games.
- Note that a cooperative outcome can evolve in a noon-cooperative game environment.

Static vs. Dynamic Games

- In static games, players are making decision at the same time, i.e. simultaneously, thus often call simultaneous games.
  - rock-paper-scissors game
  - sealed-bid auction
- In dynamic games, players are moving sequentially, thus often called sequential games.
  - chess, tic-tac-toe, Go etc.
  - Market entry
- Remarks:
  - How about a voting game?
  - **What matters is the information**, not the physical order of players' movements.

Static Games

- In static games, players make decisions without knowing what others did – thus called simultaneous movement game

- The static games are usually presented through a game matrix (often called normal form games or strategic form games) highlighting the following three key aspects of the strategic situation:
  - The players and their identities
  - The feasible strategy set for each player
  - The payoffs associated with each strategy profile

Prisoner B / Prisoner A	**Prisoner B stays silent (*cooperates*)**	**Prisoner B testifies (*defects*)**
Prisoner A stays silent (*cooperates*)	Each serves 1 year	Prisoner A: 3 years Prisoner B: goes free
Prisoner A testifies (*defects*)	Prisoner A: goes free Prisoner B: 3 years	Each serves 2 years

Prisoner's dilemma (Wikipedia)

Let’s Play the Games 1

Prisoner’s Dilemma

(Group A, Group B)		Group B: Red	Group B: Blue
Group A	Red	(2, 2)	(0, 3)
	Blue	(3, 0)	(1, 1)

- Red → Deny / 合作
- Blue → Confess / 背叛

Think about the players’ preferences

For Player A

1) A deny + B confess
2) A deny + B deny
3) A confess + B confess
4) A confess + B deny

Let's Play the Games 2

Coordination Game

(Group A, Group B)		Group B: Red	Group B: Blue
Group A	Red	(2, 2)	(0, 0)
	Blue	(0, 0)	(1, 1)

- Red → Left / Up
- Blue → Right / Down

Think about the players' preferences

For Player A

1) A Left + B Left
2) A Right + B Right
3) A Left + B Right
4) A Right + B Left

Let's Play the Games 3

Battle of Sexes

(Group A, Group B)		Group B: Red	Group B: Blue
Group A	Red	(2, 1)	(0, 0)
	Blue	(0, 0)	(1, 2)

- Red → Prize fight
- Blue → Ballet

Think about the players' preferences

For Male

1) M prize fight + F prize fight
2) M ballet deny + F ballet
3) M prize fight + F ballet
4) M ballet + F prize fight

Let's Play the Games 4

Chicken / Hawk-Dove

(Group A, Group B)		Group B: Red	Group B: Blue
Group A	Red	(0, 0)	(3, 1)
	Blue	(1, 3)	(2, 2)

- Red → Aggressive
- Blue → Yield

Think about the players' preferences

For player A

1) A aggressive + B yield
2) A yield + B yield
3) A yield + B aggressive
4) A aggressive + B yield

(Nash) Equilibrium

- An equilibrium is a stable outcome in a strategic situation.
- At equilibrium:
  - Each player has chosen the best response to what others are doing
  - No one has incentives to change their decision alone
  - The outcome may be stable, even if it is not the best for everyone
- In plain language: “I made the best choice I could, given what I expected others to do — and they did the same.”
- Why equilibrium matters
  - It helps us predict what outcome a game may settle on
  - It explains ***why some inefficient outcomes persist***
  - It gives managers a way to think more clearly about strategic interaction

Prisoner's Dilemma

What is the equilibrium?

(Group A, Group B)		Group B	
		Red	Blue
Group A	Red	(2, 2)	(0, 3)
	Blue	(3, 0)	(1, 1)

- Red → Remain silent / 合作
- Blue → Confess / 背叛

Logic

- Mutual cooperation is better for both
- But each side has an incentive to defect
- individual incentive pushes both sides toward a worse joint outcome

Coordination Game

What is the equilibrium?

(Group A, Group B)		Group B	
		Red	Blue
Group A	Red	(2, 2)	(0, 0)
	Blue	(0, 0)	(1, 1)

- Red → Left / Up
- Blue → Right / Down

Logic

- Both players want to coordinate
- The problem is choosing the same equilibrium
- There may be more than one good outcome

Battle of Sexes

What is the equilibrium?

(Group A, Group B)		Group B	
		Red	Blue
Group A	Red	(2, 1)	(0, 0)
	Blue	(0, 0)	(1, 2)

- Red → Prize fight
- Blue → Ballet

Logic

- Both prefer being together rather than mismatching
- But each prefers a different "together" outcome

Chicken / Hawk-Dove

What is the equilibrium?

(Group A, Group B)		Group B: Red	Group B: Blue
Group A	Red	(0, 0)	(3, 1)
	Blue	(1, 3)	(2, 2)

- Red → Aggressive
- Blue → Yield

Logic

- Each side wants to be tough if the other side backs down
- But if both act aggressively, both suffer

Dominant Strategy

- If a strategy gives the player better payoffs than her other strategies regardless of what other player might choose, the player would surely adopt that strategy – called a '**Dominant Strategy**'.

(Group A, Group B)		Group B	
		Red	Blue
Group A	Red	(2, 2)	(0, 3)
	Blue	(3, 0)	(1, 1)

Prisoner's Dilemma

- Red → Deny / 合作
- Blue → Confess / 背叛

Dominant Strategy

Commandment on Dominance

- If you have a dominant strategy, use it.
- Expect your opponent to use her dominant strategy if she has one.

What Makes a Prisoners' Dilemma Game?

- Each player should have her own dominant strategy.
- The (dominant strategy) game outcome is not jointly optimal. (i.e. There exist another outcome, which makes all players better off.)

Prisoners' Dilemma in Pricing Game

What is the dominant strategy?

		Firm B	
		Price Low	Price High
Firm A	Price Low	(1, 1)	(3, 0)
	Price High	(0, 3)	(2, 2)

We have seen that cooperation can increase the total payoff space. Why then don't players cooperate more often?

What should we do then?

A. The best is not to get caught in a prisoners' dilemma from the beginning.

B. Once caught, you should learn how to get out of it.

C. Repeated interaction may incentivize the players to play cooperatively.

But how?

Prisoners' Dilemma in Pricing Game

What is the dominant strategy?

		Firm B	
		Price Low	Price High
Firm A	Price Low	(1, 1)	(3, 0)
	Price High	(0, 3)	(2, 2)

?

We have seen that cooperation can increase the total payoff space. Why then don't players cooperate more often?

But how?

Change the structure of the game by making your product less sensitive to rival's prices.

The key idea is to **improve your market power**:

- Seek a competitive advantage: Differentiate your product with design improvements, service, or advertising etc. Example: Evian water
- Seek an unfilled niche; "Become a monopolist."

“I Trick You or You Trick Me”

Final Test Bonus Question: You can get free bonus marks if you (and your classmates) make right choices.

Choose 5 free extra marks or 2 marks.

- *You get whichever extra marks you chose if less than 20% of all students chose 5 marks.*
- *Otherwise, no one gets any extra marks.*

- Dominance concept does not apply to all games.
- “Dominant strategy concept” is strong if it exists. But, not all games have dominant (or dominated) strategies: the concept is not universal enough.

Game: 70% of the Average

Rules

- Each student picks **one integer from 0 to 100**.
- The winner is the student whose number is **closest to 70% of the class average**.

The game shows that in strategic situations, success depends on anticipating others, not just choosing what you personally like.

El Farol Bar Problem 酒吧博弈

- 情境设定
  - 一个小镇有 100 人
  - 每个周末，每个人都要决定：去酒吧 或待在家
  - 酒吧的最佳接待人数是 60 人
  - 超过 60 人：过于拥挤，体验变差
  - 低于 60 人：环境更舒适，去的人更享受
- 为什么会变成博弈？
  - 你的选择是否合理，取决于别人会不会去。因此，这不是一个单纯的个人决策，而是一个相互依赖的战略互动。

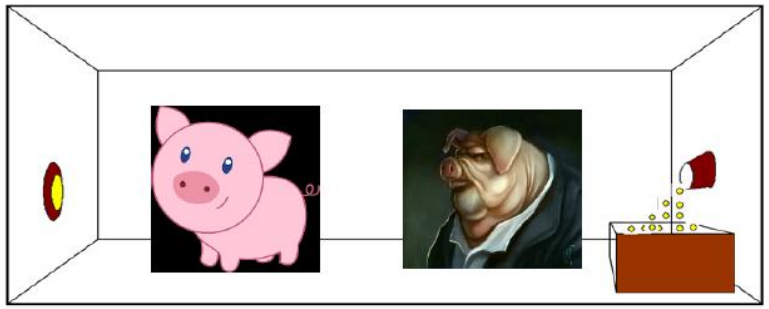
动态逻辑

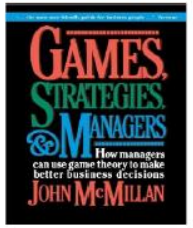
- **如果多数人都去** → 酒吧拥挤 → 下次很多人不想去
- **如果多数人都不去** → 酒吧反而舒服 → 下次更多人又想去
- 结果：系统不断波动，很难稳定在“刚好 60 人”

管理启示

- 决策不只取决于环境，也取决于**他人的预期与行动**
- 跟随多数未必最优
- 战略思维的关键是：**预判他人如何预判他人**

Rational Pigs





Source:
*'**Games, Strategies & Managers**'*
(by John McMillan)

- Two 'rational' pigs, one big (called Napoleon) and the other small (called Pinkie), are put in a cage. There is a button at one end of the cage and food dispenser at the other end which dispenses **10g** of food when button is pressed.
- Now if the pigs want food, they need to press the button; the pig that pressed the button must run to the food dispenser while the other pig has started to eat. Napoleon, being big and dominant, can eat all the food remaining once it reaches the dispenser. So Pinkie's only chance to eat is when she reaches food before Napoleon, yet still need to subordinate to Napoleon once he arrives. Pressing the button and running to the food dispenser involves calorie expenditure, which costs **2g** to each pig.

Rational Pigs

	Payoff (Napoleon's, Pinkie's)
If Pinkie presses the button, Napoleon eats all 10g	(10, -2)
If Napoleon presses the button, Pinkie can eat 7g before Napoleon arrives	(1, 7)
If both press the button, Pinkie can run faster (given Napoleon's fat shape) and eat 3g	(5, 1)
If neither presses, no food of course	(0, 0)

Question: Who is going to press the button?

Rational Pigs

	Press	Wait
Press	5, 1	1, 7
Wait	10, -2	0, 0

- So, who is going to press the button? Why?
- Lesson
  - At times, a weakness can be a strategic strength and vice versa.
  - Often quoted as ‘Judo Economics’

Business Analogy of the Pig Games: OPEC Game

- OPEC members share a common incentive to produce less, and agree on the tight production quotas.
- Saudi Arabia (SA) is a “stronger” player than the rest of OPEC due to its relatively large oil reserves and low production costs.
  - Once agreed, an individual member has a unilateral incentive to disobey the agreed quota and produce more to take advantage of the supported oil price.
  - If the rest of OPEC obeys, Saudi Arabia also has an incentive to cheat.
  - But if the rest of OPEC do not obey its cartel quota (and produced too much), it is still in the Saudi's interest to keep to its cartel quota (since SA’s share is too big and SA had better support the oil price by reducing its own production)

Takeaway: SA’s strength in market power may turn into the weakness in strategic power.

Small-scale entrants
VS.
Large-scale incumbents

Sequential Games

We now extend our analysis to include **‘dynamic’** – in the sense of **sequential** decision making process – features of the strategic decision.

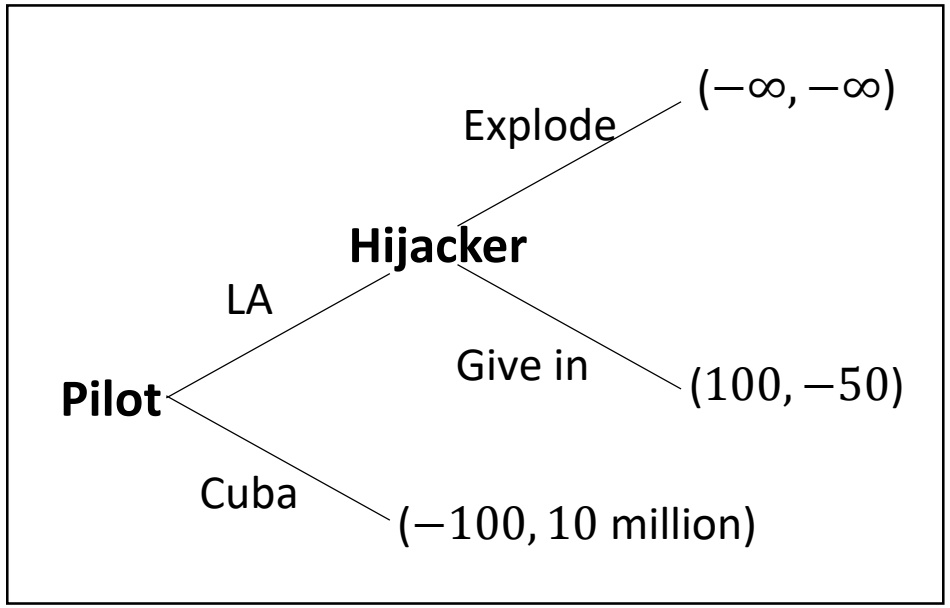
The sequential games, or extensive form games, or dynamic games, are useful to illustrate the followings:

- Different timing of strategic choices by players
- Multiple decision points for players
- Various ‘actions’ available to each player at each decision point, and the information structure for the players. (Distinguish actions and strategies!)

Sequential Games

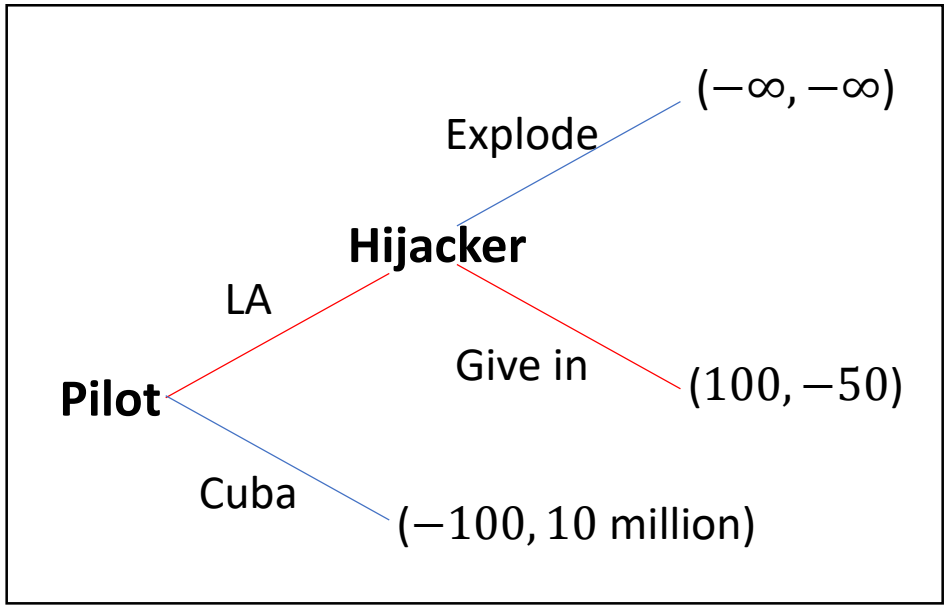
Scenario

- A hijacker wants the pilot to fly to **Cuba**
- The pilot must decide first:
  - **Comply** and fly to Cuba
  - **Refuse** and land at **LA**
- If the pilot lands at LA, the hijacker then decides whether to:
  - **Give in**
  - **Carry out the threat**



Equilibrium Concepts in Sequential Games

- Nash equilibrium concept is still valid in extensive form games.
- However, in many instances, Nash equilibrium is often multiple and not all Nash equilibriums are equally reasonable: Nash equilibrium concept has its limitations.



We find two Nash equilibriums :
(LA, Give In) and (Cuba, Explode).

Issue: Which one is more (or less) likely between the two Nash outcomes?

- (Cuba, Explode) is a Nash equilibrium but a poor predictor of how the players behave.
- Reason is: it incorporates a **non-credible threat(or empty threat)** by Hijacker.
- In conclusion, (Schiphol, Give in) makes better sense than (Cuba, Explode).

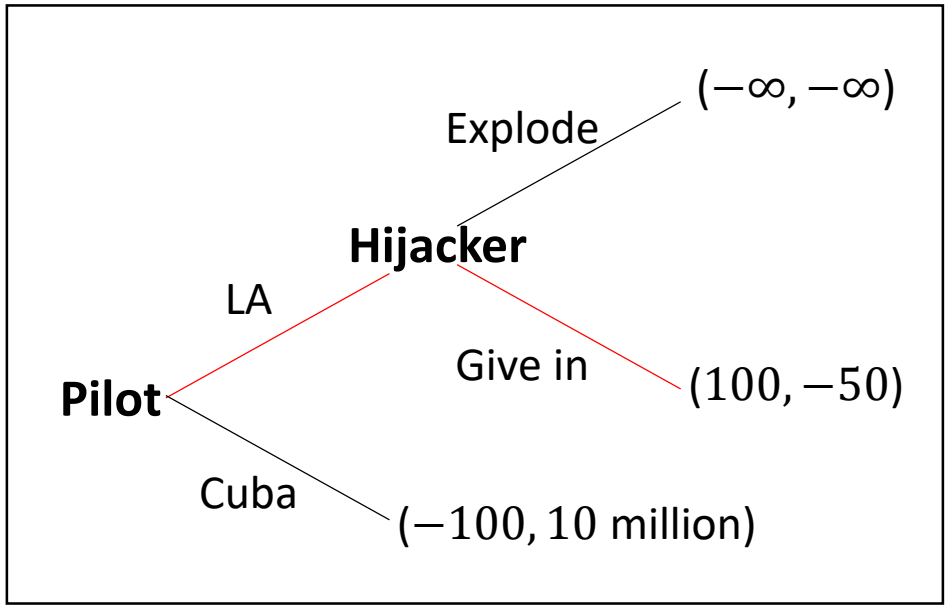
Backward Induction and Subgame Perfect Nash Equilibrium

Any algorithm then to sort out reasonable equilibrium outcomes in sequential games?

Commandment: Backward Induction

In finite dynamic games:

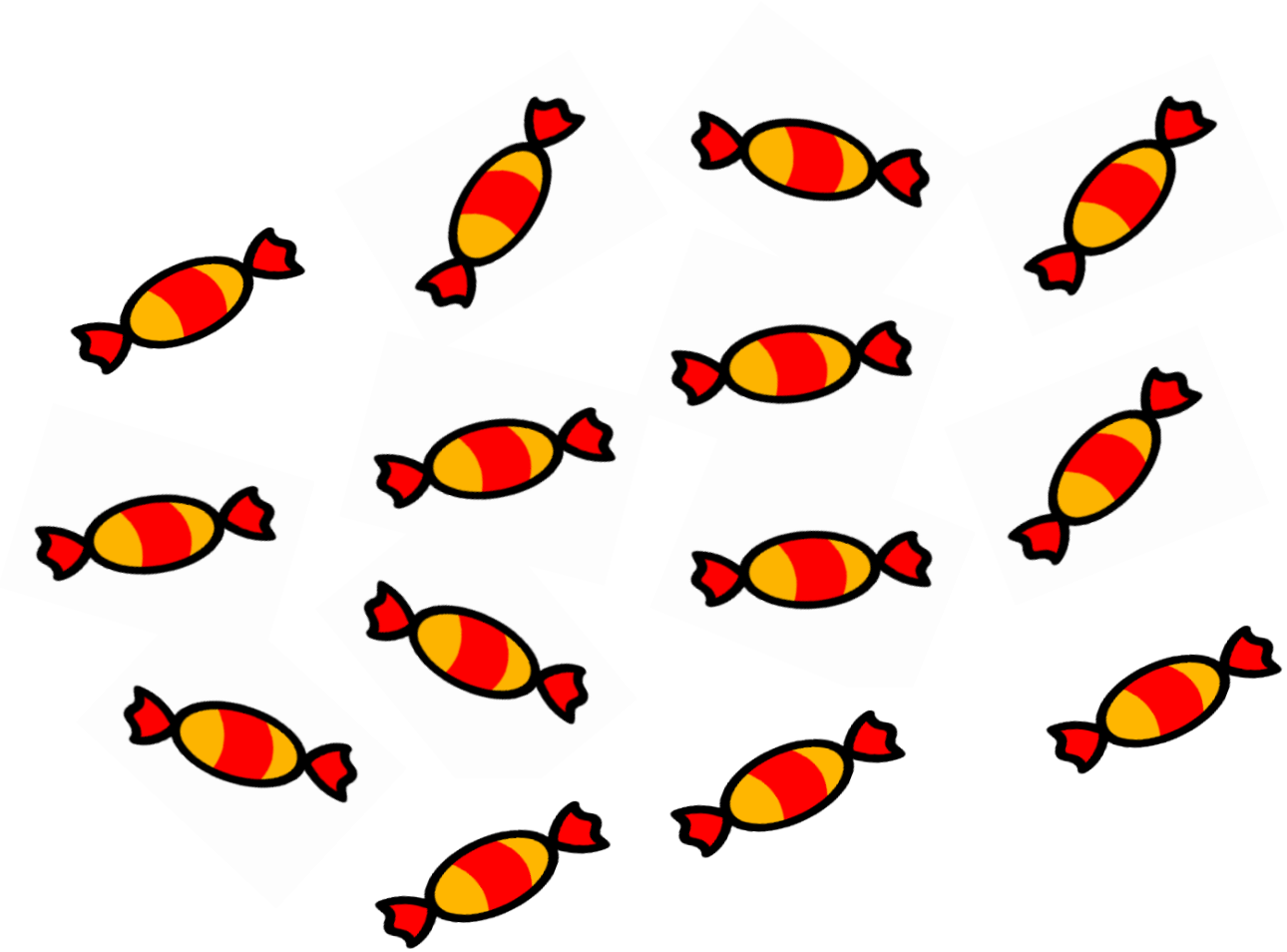
We start with the last period first and work backwards through successive nodes until we reach the beginning of the game.



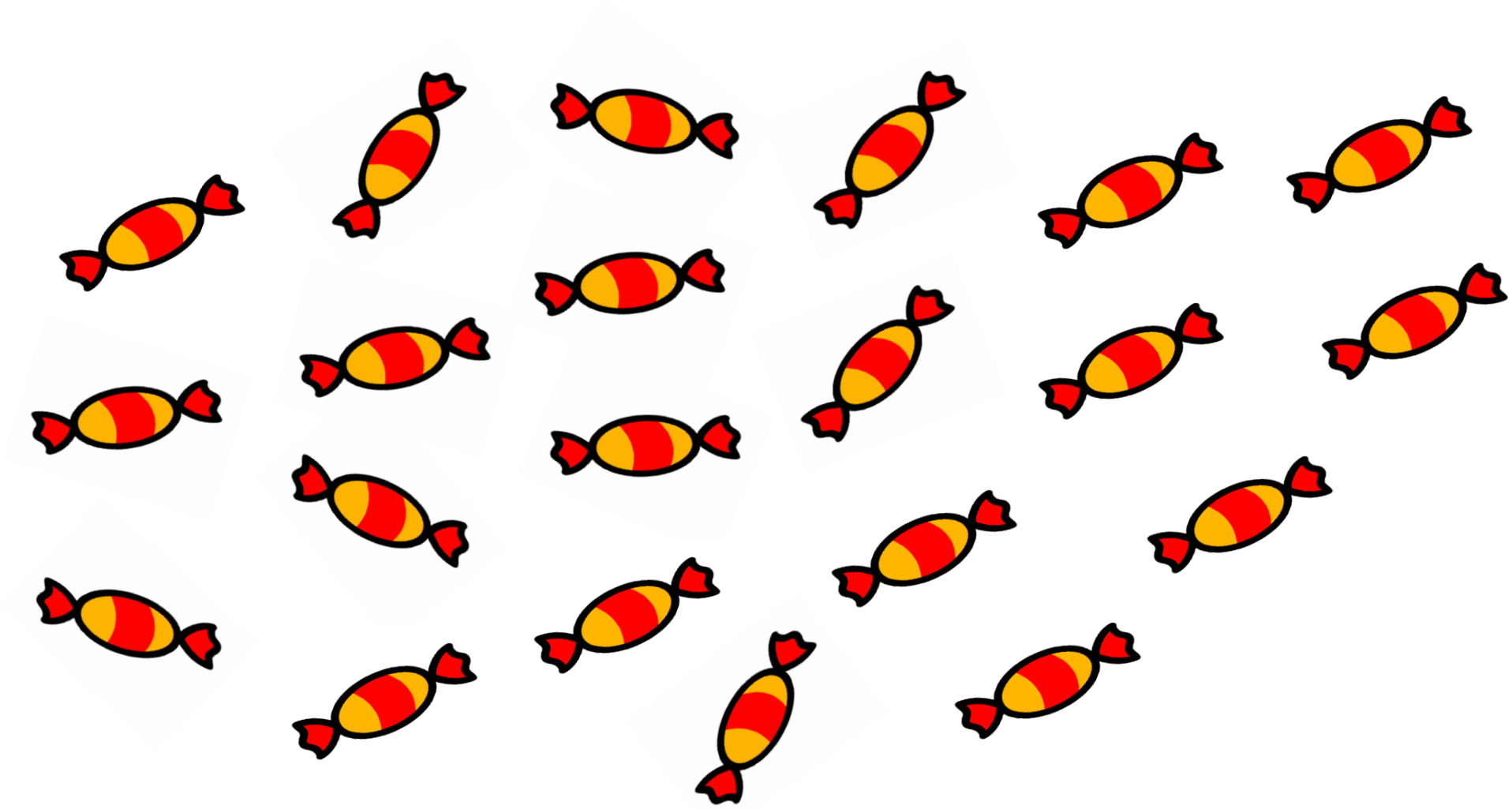
- This way, we can rule out empty threats or promise.
- (LA, Give in) is the backward induction equilibrium.

Experiment – Second Move Advantage

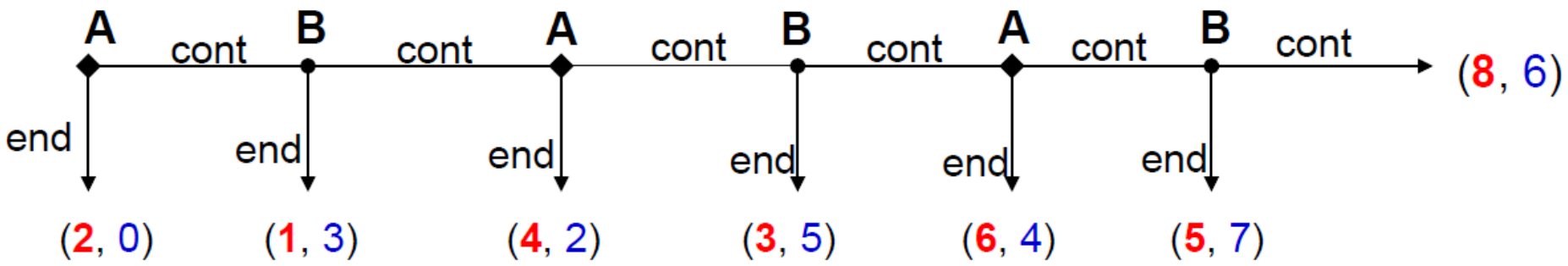
- Two players
- Each turn, a player must take 1 to 4 candies
- The player who takes the last candy wins



Experiment – First Move Advantage



Centipede Game



♦ = A's moves

• = B's moves

Payoffs: (A's, B's)

How would you play the game if you were A? How about B?

Tigers and Sheep

There are 100 hungry tigers and 1 sheep on an island.

Rules

- Tigers are perfectly rational.
- All tigers prefer:
  - eating the lamb and surviving
  - not eating the lamb
  - eating the lamb and then being eaten by another tiger
- If a tiger eats the lamb, it becomes sleepy.
- A sleepy tiger can then be eaten by another tiger.

Key question: Will any tiger eat the lamb?

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Divide the $100 or Get Nothing

Two players must divide $100.

- Step 1: Player 1 moves first
- Propose how to divide the $100
  - for example: $80 / $20, $50 / $50, $90 / $10
- Step 2: Player 2 observes the offer
  - Accept
  - Reject

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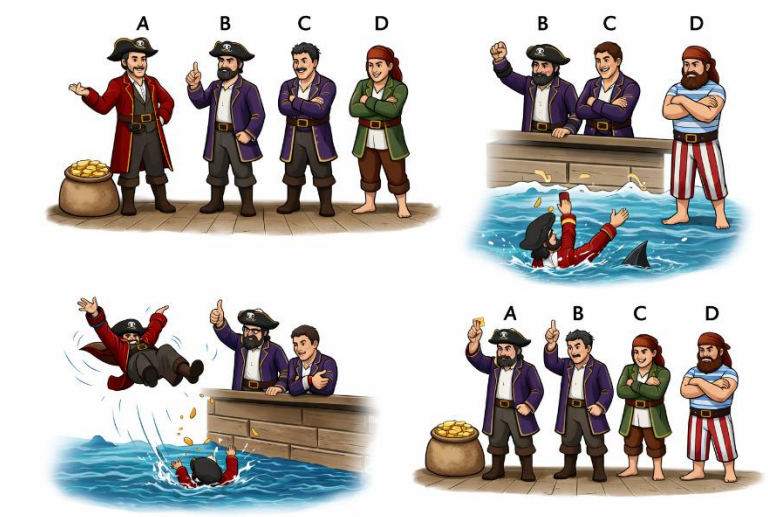
Pirate Ship 海盗分金

5 个海盗按资历从高到低排序：A, B, C, D, E
他们要分 100 枚金币，并遵守以下规则：

- 资历最高的海盗先提出分配方案
- 所有活着的海盗对方案投票
- **若赞成票≥ 50%**，方案通过，按此分配
- 若未通过，提出方案的海盗被扔下海
- 然后由下一位资历最高的海盗重新提出方案
- 所有海盗都是**完全理性、完全自利、且都知道其他人也如此**

问题：A 应该如何分，才能让自己拿到最多金币并活下来



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Rationality

Preferences

Sequence

Conflict of interest

Information structure

劣币驱逐良币，二手车市场

- Are you rational?
- Is your opponent rational?
- What information do you have?
- What information does your opponent have?
- Trust / commitment / coordination?
- Do you have conflict of interest?